Guidelines for Preparing Pump CFD Model from CAD model

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1. Purpose

The document describes a guideline for preparing the CAD model of a generic pump (single or multistage type), adapted and set up to be imported into a CFD environment.

This procedure is propaedeutic for the SOP-504 "Guidelines for Meshing Pump (stage geometry) for CFD Analysis".

2. Scope

This procedure is applicable to any centrifugal pumps, single or multistage, single or double entry, with volute or diffuser discharge for the Nova Milanese and La Spezia plants.

3. Abbreviations

CFD		*Computational Fluid Dynamics*
D1	*[mm or in]*	*impeller eye diameter*
D2	*[mm or in]*	*maximum external diameter of the impeller*
Dh	*[mm or in]*	*shaft diameter at impeller*
Eta	*[%]*	*efficiency*
H	*[m]*	*head*
LE		*leading edge of impeller blades*
Nq or Ns		*specific speed of the pump*
PS		*pressure side of impeller blades*
Q	*[m^3/h]*	*flowrate*
SS		*suction side of impeller blades*
TE		*trailing edge of impeller blades*

4. General procedure: description

The guideline provides general instructions to prepare a CAD model of pump interiors starting from a general CAD (typically assembly model), that will be used for CFD analysis.

CFD simulations need to model the volume of fluid flowing within a pump so the 3d fluid domain must be extracted from the pump assembly.

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Usually, the main components that make up the fluid volume assembly of a pump are listed below:

- suction casing
- impeller
- discharge casing (volute and or diffuser)
- crossover/return channel
- secondary flow lines
- leakage domain between impeller and casing

A combination of these components allows to realize the type of hydraulics of the selected pump; in the Annex A the pump types according to the API 610 standardization are reported [5].

This operation of extracting the fluid domain from the solid CAD of the pump is the first step taken to prepare the CFD simulation activity.

The guideline for preparing the CAD model of impeller is not covered by this document, as it is already discussed extensively in SOP-481.

A general discussion is presented below, with the aim to provide the user with the necessary information and recommended criteria to make a CAD model of pump using general purpose tools CAD 3D (most popular are for example Solid Works, Solid Edge, PTC CREO, AutoCAD 3D).

Depending on the requested level of accuracy of the CFD study, different type of assembly shall be selected. A list of the typical assembly used on CFD applications are shown below:

Commentato [ER1]: LDF
Qua metterei le tipologie di modelli cad da preparare per le diverse analisi che effettuiamo, con il diverso livello di accuratezza… magari una tabella potrebbe aiutare

- High fidelity:
  - complete pump (CP)
  - complete pump without balancing line (CPnobl)
- Simplified model:
  - half pump with respect to a symmetrical plane (SYM)
  - only first stage for NPSH simulation (1ST)
  - two next stages of multistage pump (STnext)
  - last stage and discharge of multistage pump (STlast)
  - suction + passage of impeller of first stage (SI)
- single component:
  - only suction (OS)
  - only passage of impeller of first/next stage (OI)

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- only volute (OV)
- only diffuser (OD)

Table 1 below correlates the model topology with the type of CFD analysis:

	-Thrust calculation, -Balance mass flow rate calculation	**Calculation of pump performance**	**For activity of redesign**	**NPSH calculation**
CP	X	X		
CPnobl		X	X	
SYM			X	
1ST				X
STnext			X	
STlast			X	
SI			X	
OS			X	
OI			X	
OV			X	
OD			X	

Table 1 : model topology and type of CFD analysis

Following Table 2 instead associates the type of pump with the topology of pump model:

	Topology of model pump		
Type of pump	**High fidelity (complete pump)**	**Simplified model**	**Single component**
OH1,OH2,OH3,OH4,OH5,OH6	CP,CPnobl		
BB1, BB2	CP,CPnobl	SYM	
BB3,BB4,BB5	CP,CPnobl	1ST,STnext,STlast	OS,OI,OV,OD

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VS1	CP	1ST,STnext	OS,OI,OD
VS2,VS3	CP		
VS4,VS5	CP		
VS6	CP	1ST,STnext	OS,OI,OD
VS7	CP		

Table 2 : Type of pump with topology of pump model

It is important to specify that the complete pump models must always be prepared in the initial phase of the activity for two reasons:

- Initial CP, for the CFD comparison with the TESTS carried out in the test center

- Final verification CP, to confirm the improvement in performance of the redesign activity

While the simplified models and the individual components are used in the intermediate phase of the activity to reduce calculation times; except for OS, the other models need appropriate boundary conditions extrapolated from the CP model.

The only model that does not need a verification with CP is 1ST for the NPSH calculation activity.

Instead, for the redesign activity of the BB1 and BB2 pumps with staggered blades impellers, it is always convenient to start with the simplified SYM model and then conclude the activity with the CP model so as to be able to verify the reduction in radial thrusts.

In VS6 and VS7 pumps with barrel, it is always suggested to model also the fluid inside the tank.

For the verification and calculation activities of the thrusts, only the CP model is prepared.

In the following figures a list of main geometry of topology of pump model:

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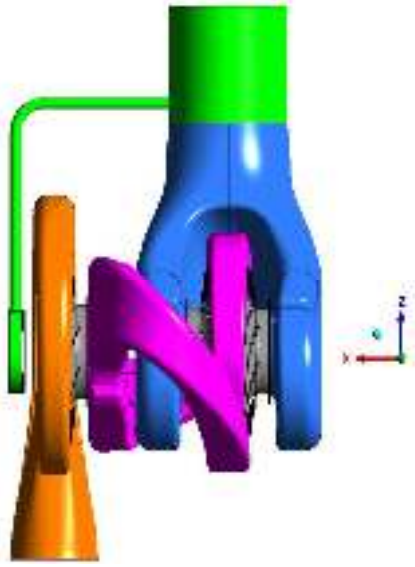
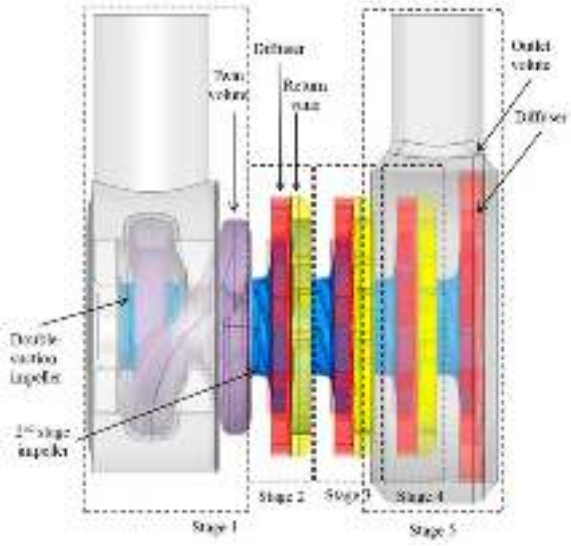
- CP (BB2)



- CPnobl (BB4,BB5)



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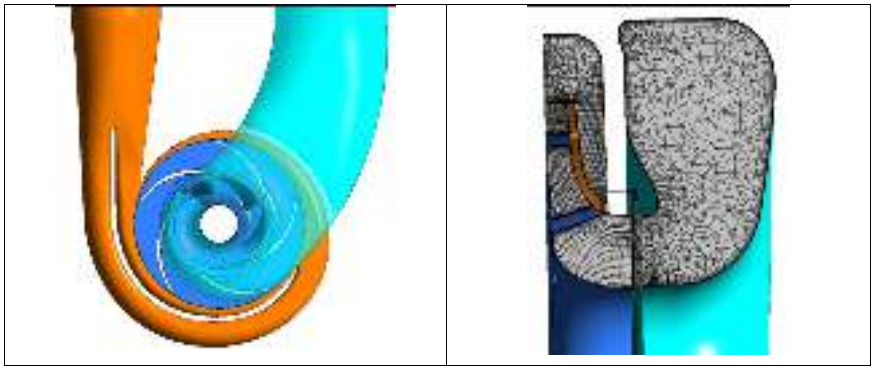
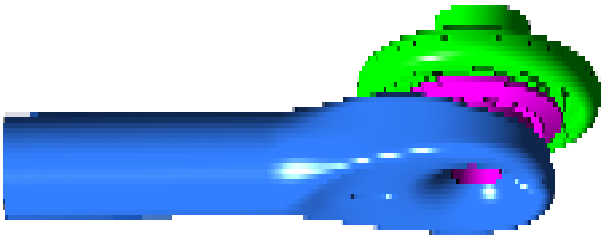
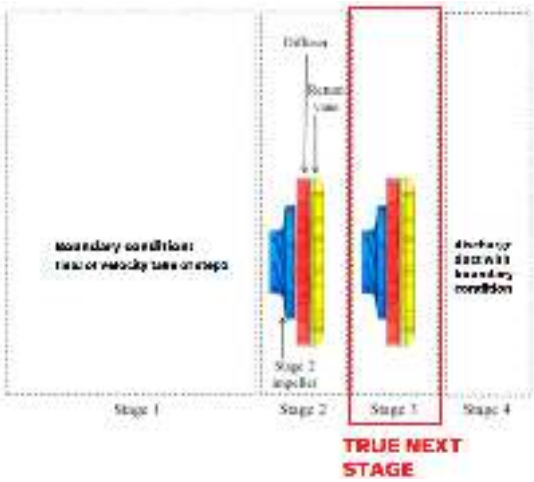
- SYM (BB1)



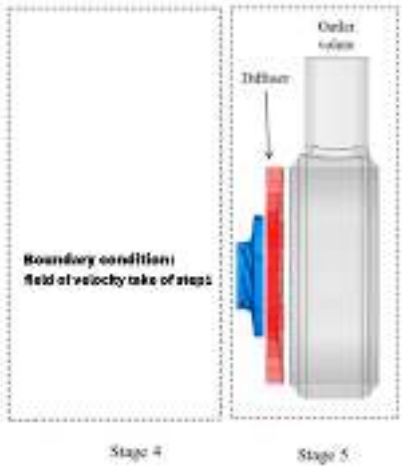
- 1ST (BB4,BB5)



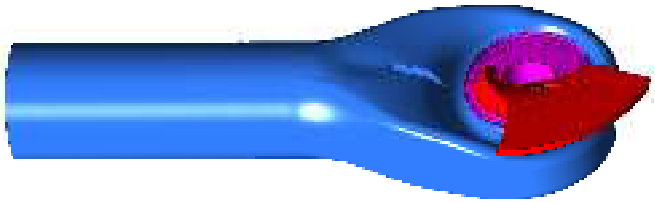
- STnext (BB4,BB5)



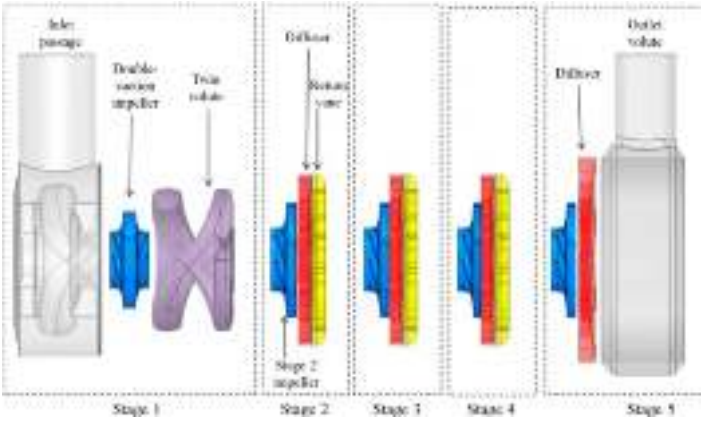
- STlast (BB4,BB5)



- SI (BB4,BB5)



- Single component (BB4,BB5)



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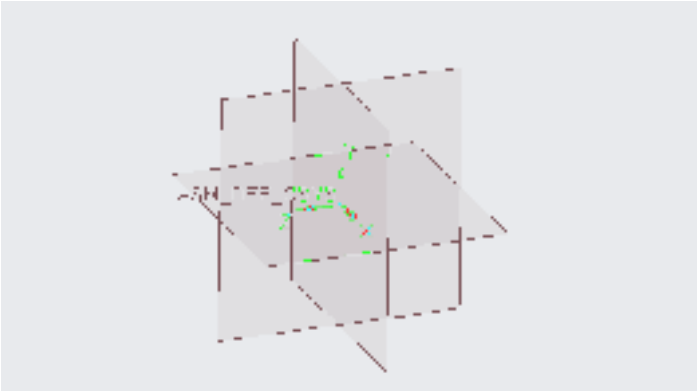
1. Assembly 3D solid CAD

The assembly phase is very important because it defines the relative position of the different components and how they are connected to each other.

In fact, before manually loading the parts that make up an assembly, it is necessary to create an orthonormal coordinate system and its reference planes.



The origin of the coordinate system is always located on the system of the rotation axis of the rotor group, and its position along the axial direction usually coincides with the direction that passes through the center of the suction duct (usually circular in shape).

To connect and constrain the different components you can use commands such as [4]:

- **Distance**—Offsets the component reference from the assembly reference.
- **Angle Offset**—Positions the component at an angle to the assembly reference.
- **Parallel**—Orients the component reference parallel to the assembly reference.
- **Coincident**—Positions the components reference coincident with the assembly reference.
- **Normal**—Positions the component reference normal to the assembly reference.
- **Coplanar**—Positions the component reference coplanar to the assembly reference.

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- **Centered**—Centers the component reference and the assembly reference.
- **Tangent**—Positions two references of different types so that they face each other. The point of contact is a tangent.
- **Fix**—Fixes the current location of a component that was moved or packaged.
- **Default**—Aligns the component coordinate system with the default assembly coordinate system.
- **Automatic**—Shows available constraints in the list after you choose a reference.

In general, it is sufficient to use the first four commands listed above to be able to create an assembly.

The necessary conditions for these commands to be used is that the contact surfaces between the components are of the Boolean type, that is:

- flat surfaces must be effectively planar

- cylindrical surfaces must be circular in shape so that the geometric axis of the surface can be created.

In Figure 1 a correct example of assembly of a BB1 pump.

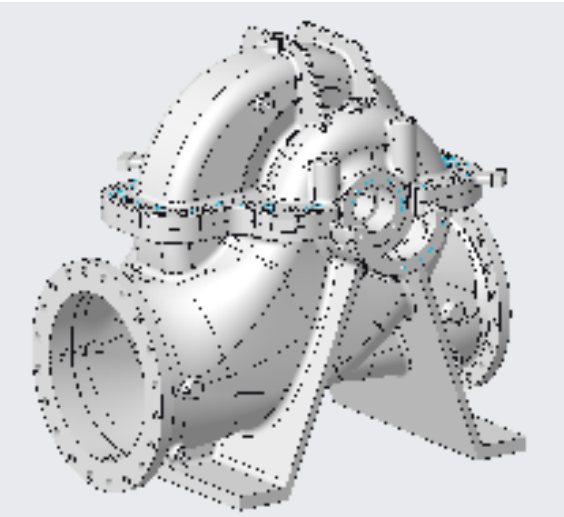

Figure 1 : example of CAD assembly, BB1 pump

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2. Extract the wet surface

In order to create the different fluid domains, the first step to follow is the selection of the wetted surfaces, since in fact it is the external surface of the fluid domain that we want to create.

In the example shown in Figure 2 all the green surfaces are part of the suction duct; as you can see, not all of them are part of the same 3D solid and often it must be searched for among multiple elements that make up the assembly; in the case chosen as example, the suction duct is located one part in the casing and the other part in the cover.

The final extracted surface is shown in the bottom image of Figure 2.

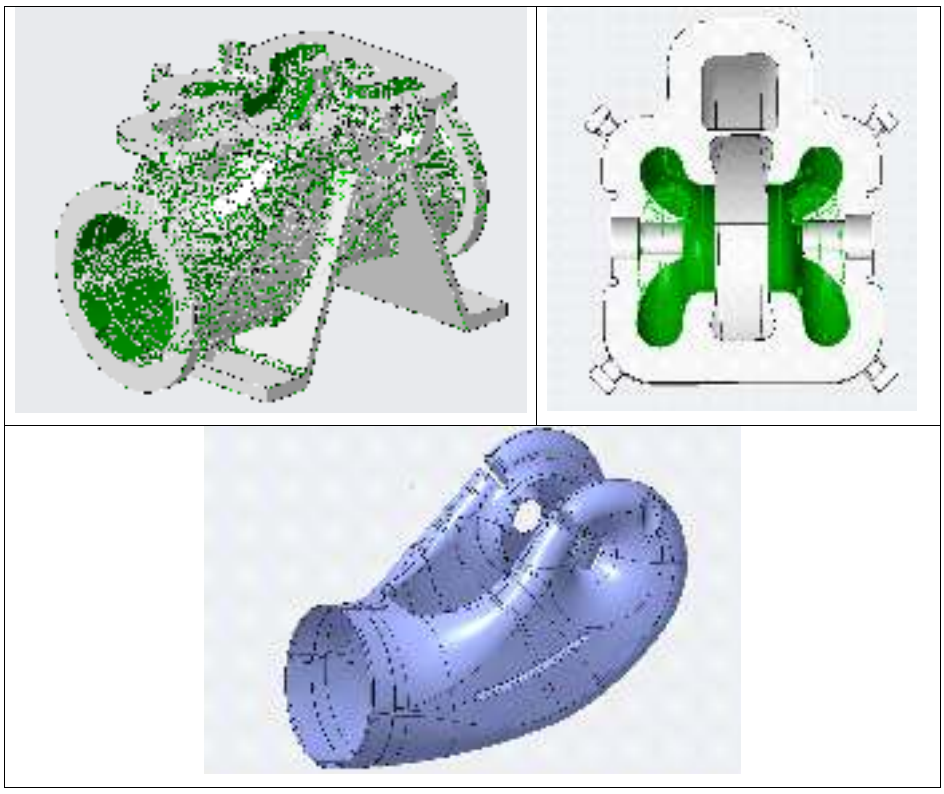

Figure 2 : example extracting wet surface of BB1 suction

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3. Solidify the domain fluid

The CFD model must operate with a closed volume of fluid.

For the solidification phase it is necessary to close the open surfaces where the fluid flows inside.

Often, pump drawings also include holes that "pierce" the surface; these are made to insert measuring instruments for reading the static pressure.

All these surfaces must be closed to create the solid.

4. Assembly 3D fluid CAD

Once all the solidified fluid domains are available you need to create a fluid domain assembly.

This operation is simple to carry out because each domain maintains its actual position with respect to the other fluid domains.

Commentato [ER2]: LDF
Potresti mettere una tabella di suddivisione dei domini per ogni tipologia di pompa e per ogni tipologia di analisi da effettuare

The following Table 3 and Table 4 show the different fluid components on a 3-stage BB5 pump.

In the 3D cad assembly, it is also necessary to draw the leakage path (see Table 4)

Suction	Diffuser of first stage	Diffuser of second stage	Diffuser of third stage	Discharge

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Table 3 : fluid domain BB5 pump 3 stages

3D cad Assembly

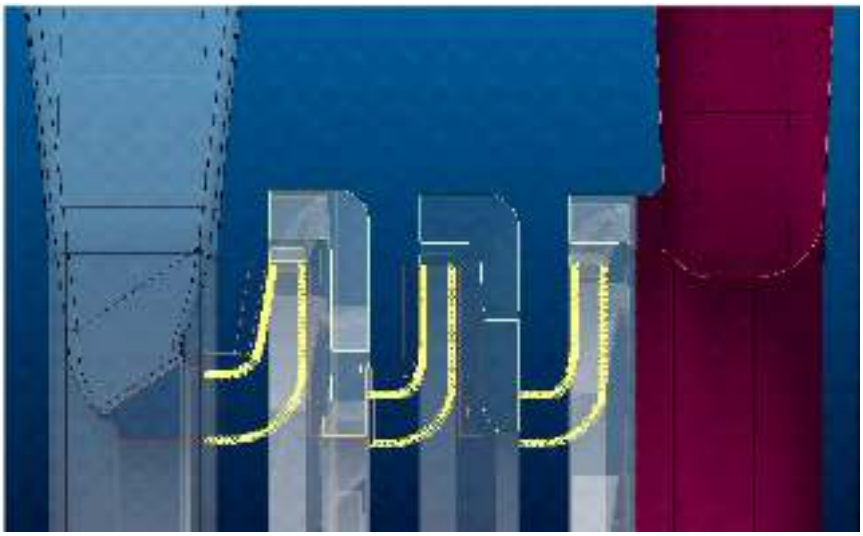

Table 4 : fluid domain assembly of 3-stage BB5 pump

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5. Level of geometric accuracy

Commentato [ER3]: Le approssimazioni dipendono dal tipo di analisi che si fanno e dalle tempistiche tipiche richieste.. Metterei il caso nostro e quello solitamente usato da GAB

The importance of geometric detail in the solid geometry of the pump very often does not have the same value as geometric detail in the fluid domain.

In other words, very often it is possible to neglect some mechanical geometric details because by experience it is known that they do not significantly influence the CFD simulation results. These are only a complication in the subsequent meshing phase.

Following Figure 3 is an example of a a BB5 pump design study drawing:

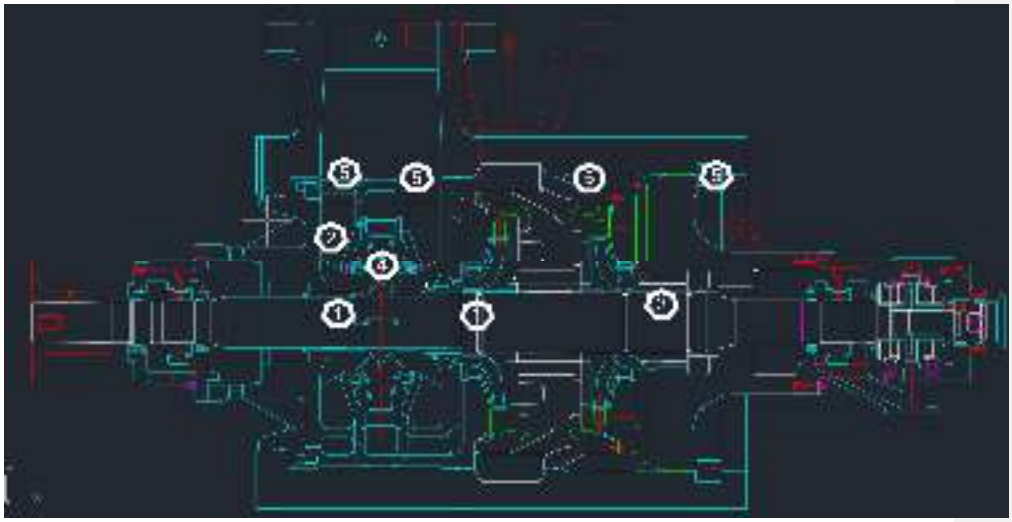

Figure 3 : example of BB5 pump design study drawing

Below Table 5 shows some geometric details that can be neglected in the simplification of the geometry, without resulting in a significant modification of the hydraulic performance results.

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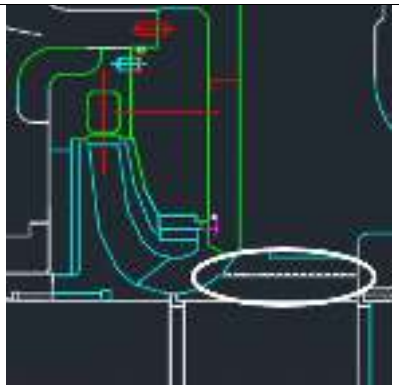
	Detail number Figure3	
1		remove the step extending the curvature of the suction duct up to the impeller inlet.
2		Simplify the leakage regions between the impeller disk and the casing, for example by using straight sections.
3		It is often not convenient to simulate the flow line but it is sufficient to impose a flow boundary condition obtained from a preliminary calculation in the CFD model (see CP and CPnobl model)

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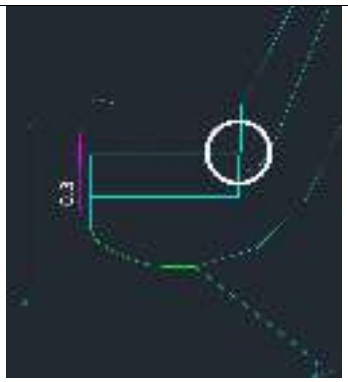
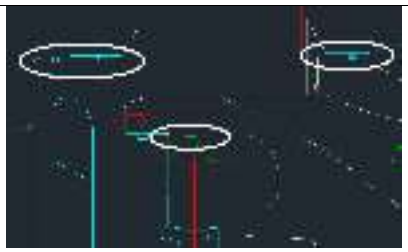
4		Simplify the leakage regions by neglecting the mismatch between the two wear rings
5		Simplify the junction regions between solid components, any edges or recesses of the fluid must be eliminated

Table 5 : detail for geometric simplification

On the other hand, there are geometric details that are crucial to have a successful simulation; for these particular details it is necessary to have an accurate geometric reconstruction of the CAD and in the subsequent meshing phase a good, localized refinement of the mesh (see SOP-490).

Some examples of geometric details to which it is important to pay more attention to are:

-shape of the blade and tongue volute (LE,TE in particular)
-shape of meridional channel of impeller, diffuser and return channel
-dimension of leakage (GAP and diametral GAP)
-shape of inlet and outlet of impeller, diffuser and volute
-diameter of flange suction and discharge

6. Balancing line

Commentato [ER4]: LDF
Indica quando va messa, come va modellata e quando può essere approssimata

The balancing line must be modeled carefully in all the following cases where:

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- the flow rate in the line is not negligible and affects the pump performance. Low specific speed pump (Nq<40) falls into this statement

- measurements are taken on the axial and radial thrusts

7. Definition of interface

The division of fluid domains is an important operation because it defines and distinguishes the different fluid domains which will communicate with specific interfaces.

The reasons why it is necessary to divide the fluid domains are several, they are the following:

- the fluid domains are both stationary and rotating so a physical separation between the stator and rotor domains is therefore necessary.

- it is convenient to create different types of meshes with different meshing tools because each component has its own geometric complexity and a different mesh definition level

- subdividing and distinguishing the different pump components allows you to estimate its performance individually and allows you to carry out a possible flexible and targeted redesign activity of the individual component.

The choice of how and where to position the interfaces constrains and limits the space available for each individual domain for the creation of each mesh.

The leakage domain has some interesting features listed here:

- it is an axisymmetric domain

- it is flexible, as it easily adapts to the needs of other domains

Figure 4 shows the design study of a BB5 pump next stage; the red dotted lines indicate the chosen boundary positions of different fluid domain; in this example it has been identified five interfaces, listed here:

- the interface between the return channel domain and the leakage domain

- the interface between the leakage domain and the impeller domain

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- the interface between the impeller domain and the leakage domain

- the interface between the leakage domain and the diffuser domain

- the interface between the leakage domain of the next stage1 and the leakage domain of the next stage2.

The leakage domain becomes the domain that allows all the different fluid domains to communicate at a fluid dynamic level; it is therefore the one that presents the greatest number of interfaces in the pump model.

Commentato [ER5]: LDF
Cerca di rendere il documento impersonale.. Formule come quella usata in questo punto possono essere elminate

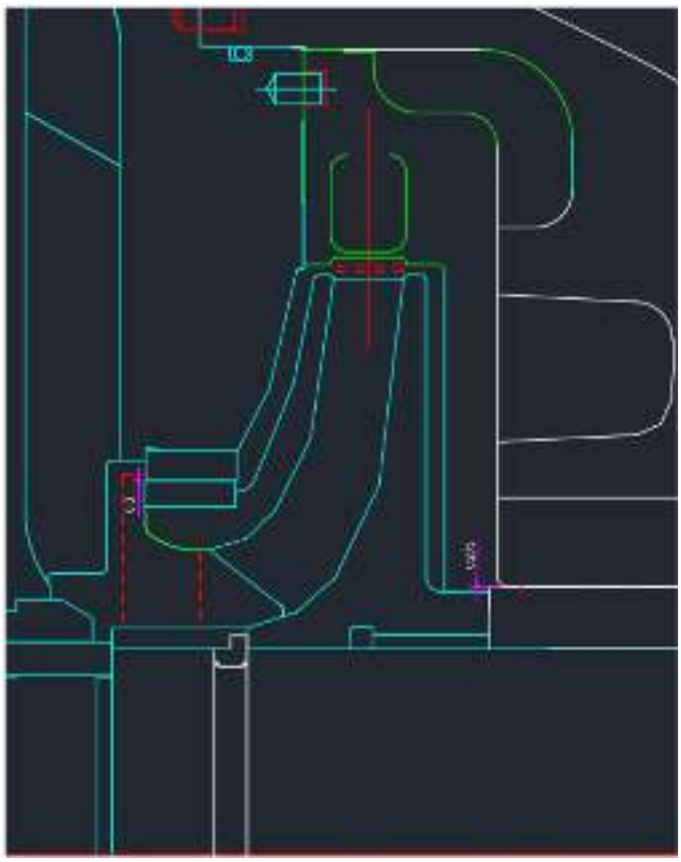

Figure 4 : example position of interfaces

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8. Preparation and design of 2D/3D leakage

Since the leakage domain is axisymmetric, it is created starting from a sketch.

The fundamental shares that characterize it are shown in table 6; the example shown in Figure 5 helps understanding these parameters.

Parameter name [mm]	Description	Example value Figure5 [mm]
D1	Inlet diameter at impeller tip	77.495*2
Dhub	Inlet diameter impeller at hub	54.5*2
b1	Meridional channel inlet of impeller	22.995
X2h	Distance between impeller hub at D2 and interface leakage-impeller	45
Lwr gapR Dwr	Wear Ring Length Wear Ring Radial GAP Wear Ring Diameter	17 0.24 (77.4495+10+5)*2
gapR_front Dwr_front		0.19 57*2
gapR_rear Dwr_rear		0.19 57*2
D2+(0.5*2)	Impeller outlet diameter + (0.5*2)	(165*2)+(0.5*2)
b2	Impeller meridional channel outlet	20
D3/Dcw-(2.7*2)	Diameter at LE of diffuser blade or volute tongue – (2.7*2)	(170*2) – (2.7*2)
b3	Diffuser or volute meridional channel inlet	30

Table 6 : main parameters for preparation of leakage

As can be seen from Table 4, the parameters D2, D3 and Dcw need an additional or subtractive dimension to take into account the passage of the mesh around the impeller blades and the diffuser blades (in the same way also around the tongue of the volute).

An example that shows the reasons for this caution in creating the leakage is shown in Figure 6.

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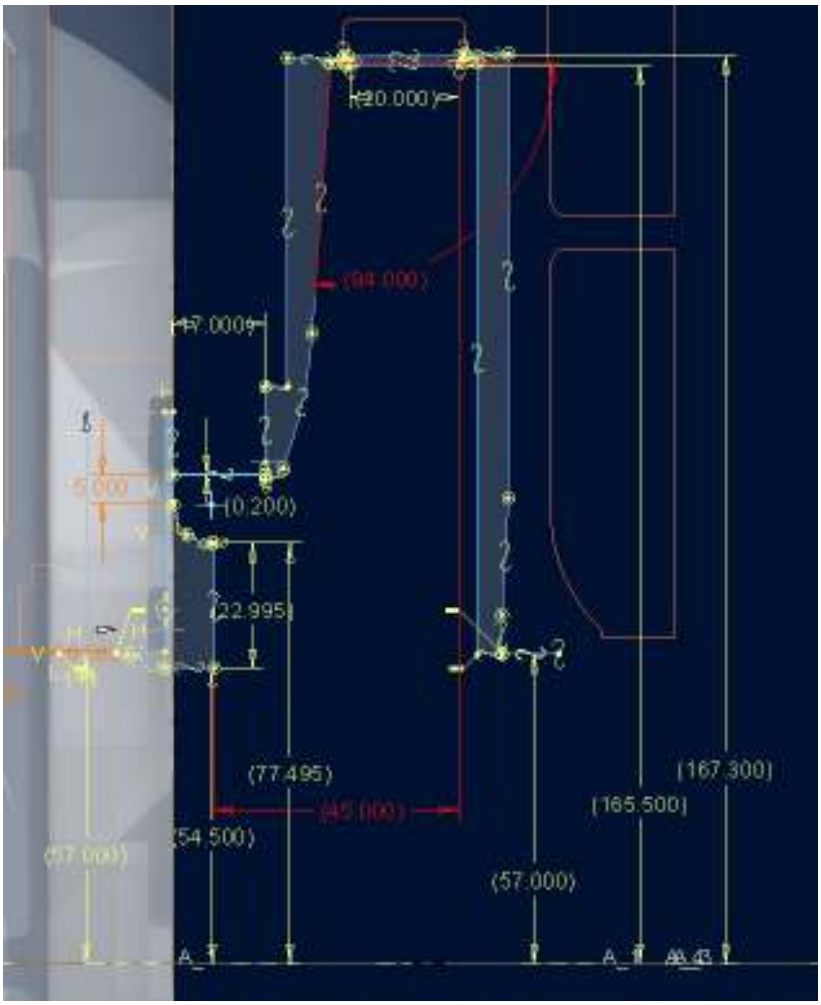

Figure 5 : example of 2D sketch for preparation of leakage domain

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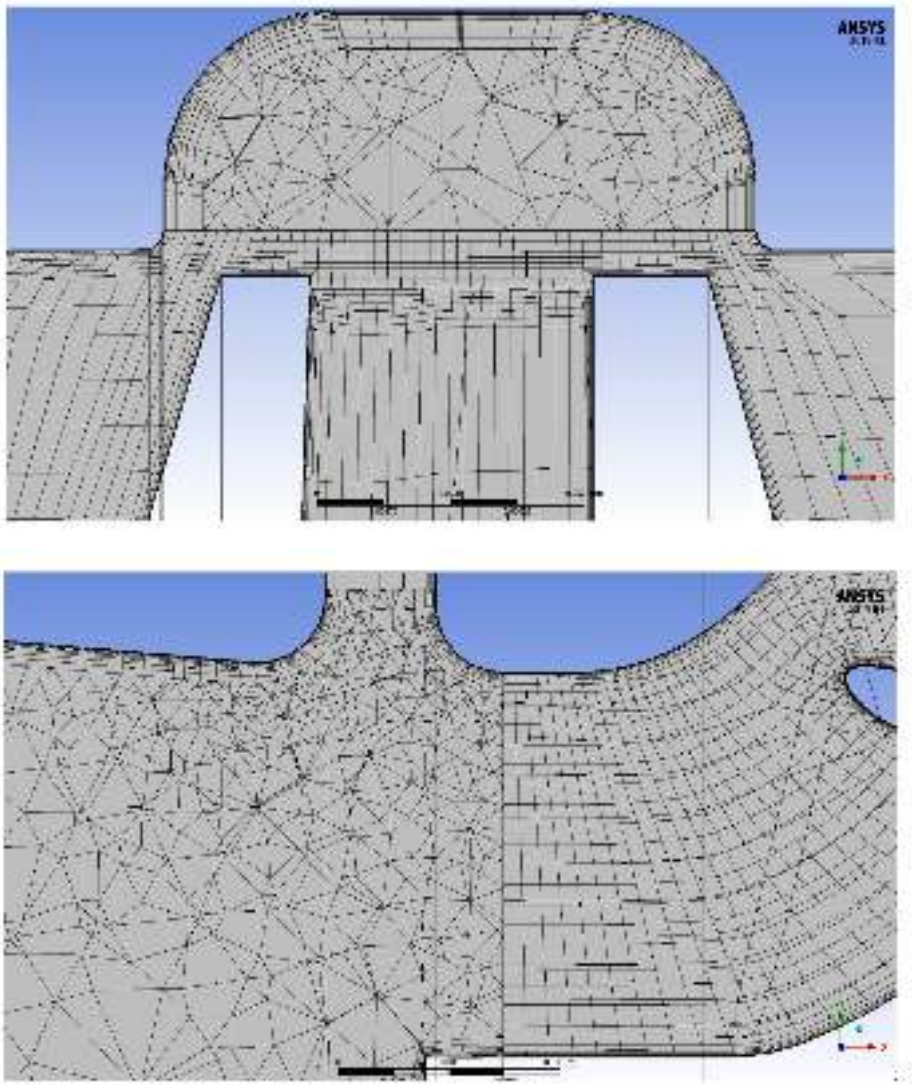

Figure 6 : example of 3D mesh section of leakage

The GAPR size of the leakage is governed by the table in Figure 7, extracted from the API 610 standards *[5]*.

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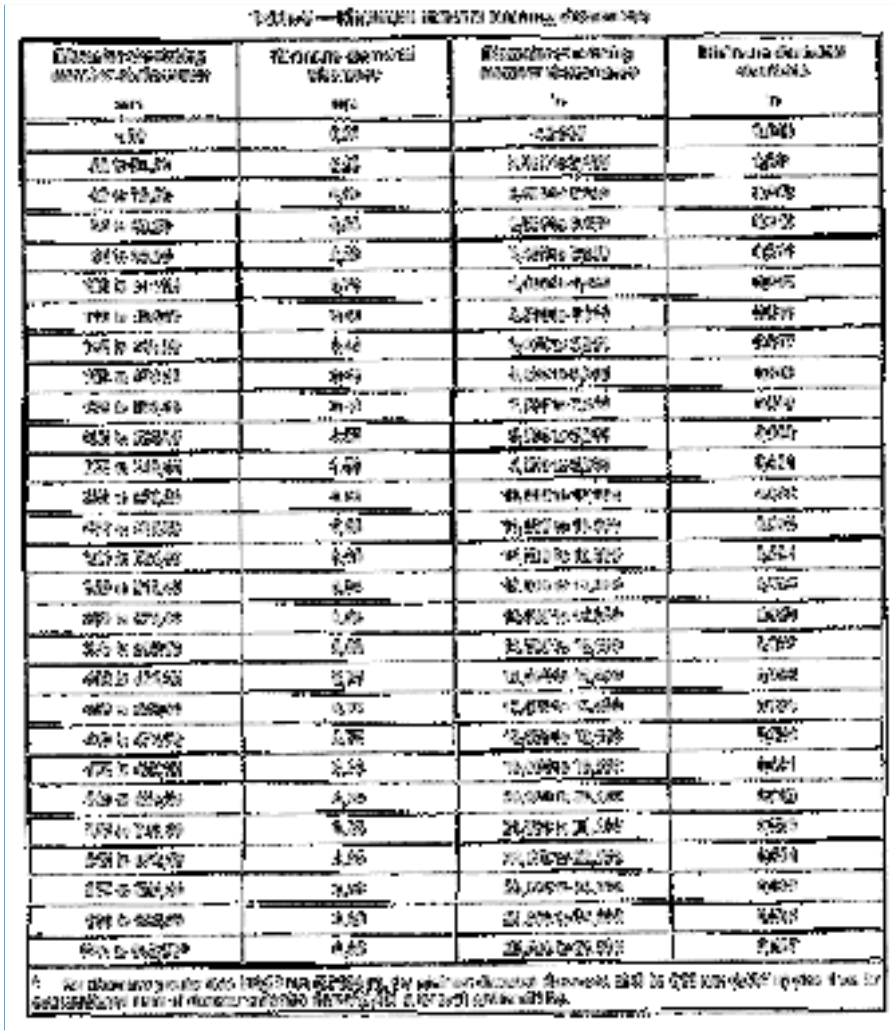

Figure 7 : Minimum internal running clearances, API 610

Commentato [ER6]: LDF
Non lo metterei, fa parte del dimensionamento del leakage, non della preparazione del cad per la cfd

Figure 8 and Figure 9 show two examples of leakage generation, for a BB5 pump and for an OH pump.

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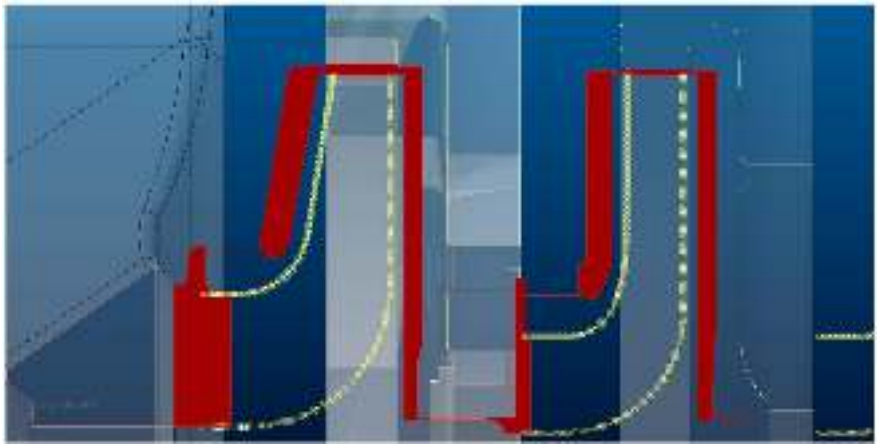

Figure 8 : example of 2D leakage creation,BB5 pump

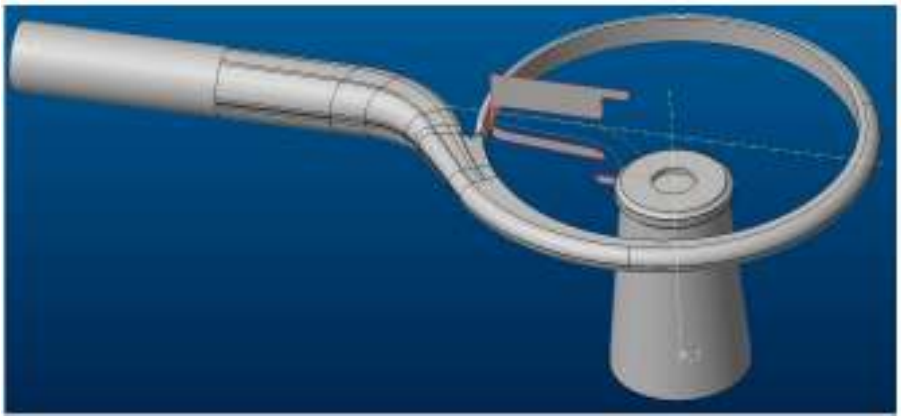

Figure 9 : example of 2D leakage creation, OH pump

It is important to point out that CFD simulation always needs 3D domains, it is therefore necessary to generate a 360° leakage from the leakage sketch (or an angular fraction of it if you want to use its rotational periodicity, and thus reducing the number of elements used in this mesh).

Commentato [ER7]: LDF
Direi che il calcolo CFD considera un volume di leakage che occupa 360° o meno. Non viene realizzato dal cad, ma il modello CFD lo considera. Metti anche quello approssimato da GAB

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9. Neutral output formats

Once you have obtained the 2D leakage you can decide to proceed in the following ways:

- extraction of the 2D leakage, then use of ICEM mesh tool which allows you to create a 3D revolution starting from a structured 2D mesh

- extraction of the 3D leakage obtained as a solid revolution within the CAD; with this geometry you can create a structured or unstructured 3D mesh by a generic meshing tool.

Typical formats required as input by mesh tools are the following:
- .iges
- .step
An example of input files and related tool of meshing is in the following table:

Component type	Geometry	File extension	Meshing tool type
suction	3D	.iges/.step	ANSYS MESHING
leakage	2D	.iges	ICEM
Impeller, bowl	3D	.iges/.step	TURBOGRID
diffuser, cross-over, return channel	3D	.iges/.step	ANSYS MESHING
discharge	3D	.iges/.step	ANSYS MESHING
pipe suction, pipe discharge	3D	.iges/.step	ICEM

Table 7 : input geometry file for meshing

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5. Acceptance Criteria

This procedure is applicable for any geometry of suction, cross-over, volute and diffuser (see Table 8, Table 9, Table 10, Table 11).

In the following tables the different typologies of static component which constitute a centrifugal pump are described:

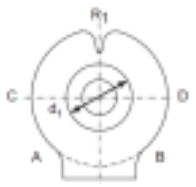
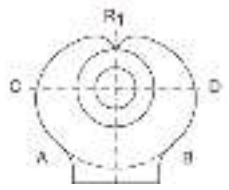
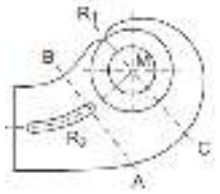
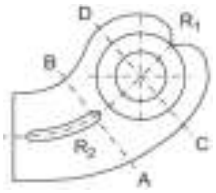
Type suction casings			
Barrel pump (radial split)		Double suction (axial split)	
Anulus	Spiral shaped	Asymmetrical	Quasi-symmetrical
			

Table 8 : Type of suction geometry [1]

Short and long cross-over for multistage	
Short Cross-over	Long Cross-over

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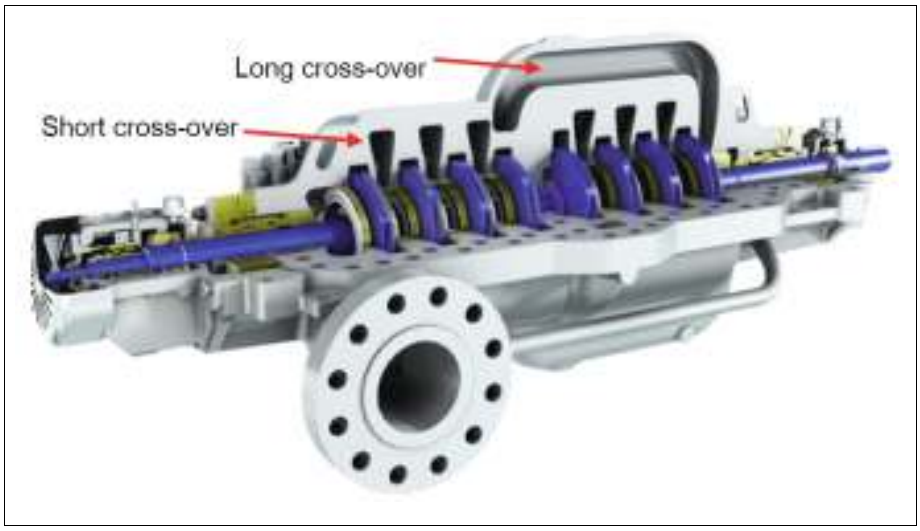


Table 9 : Type of cross-over geometry [1]

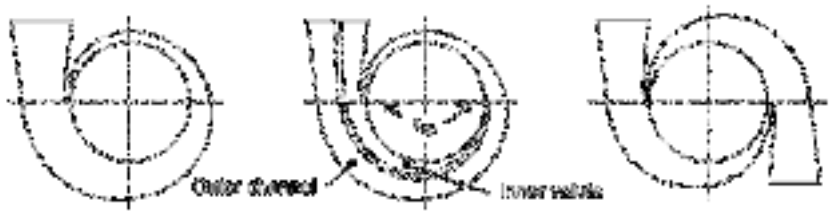
Type volute casings		
Single volute	Double volute	Twin volute
		

Table 10 : Type of volute geometry [1]

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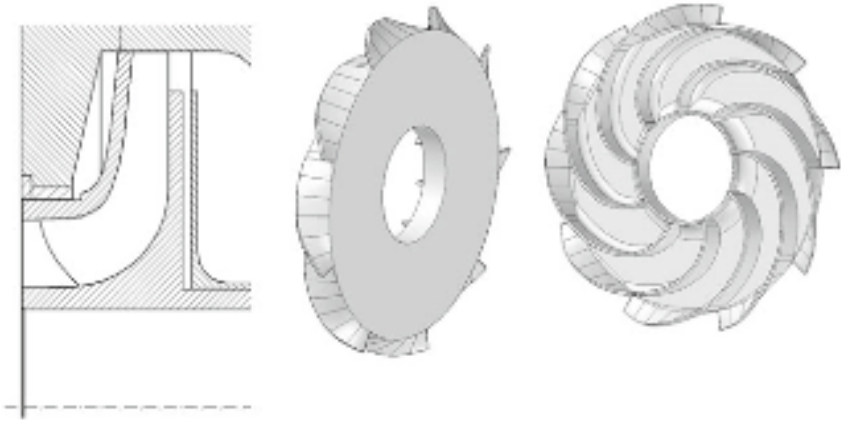
Type of diffusers casings (separated diffusers and return vanes)			
Diffuser with volute	Diffuser and annular casing as last stage	Overflow from diffuser to return vanes through vaneless annuls	Lateral overflow from diffuser to return vanes
Type of diffusers casings (continuous diffusers and return vanes)			
			

Table 11 : Type of diffuser geometry [1]

Considering the type of pump it will be necessary to work on different composition of this part of pump.

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6. References

[1] Gulich "Centrifugal Pumps", 4th.ed.

[2] Ansys User Manual

[3] PTC Creo User Manual:

[4]https://support.ptc.com/help/creo/creo_pma/r10.0/usascii/index.html#page/assembly/asm/About_the_Component_Placement_User_Interface.html

[5] API STANDARD 610, 12th Edition, January 2021

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Annex A: Pump type (*API 610 Standard)*

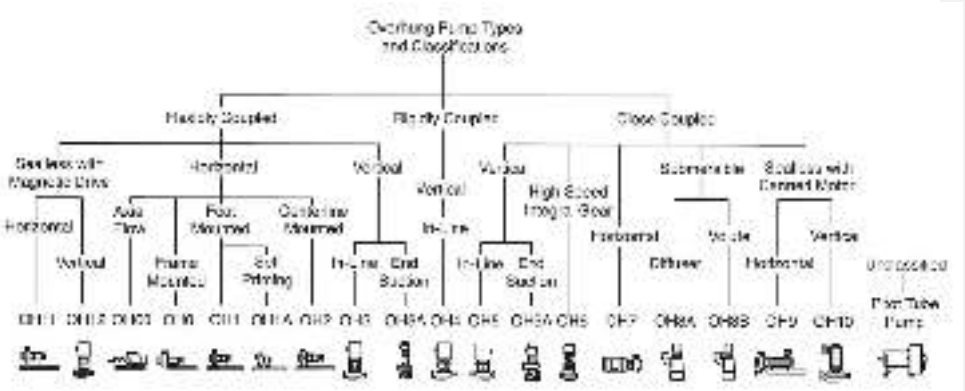

Figure 10 : Overhung pumps

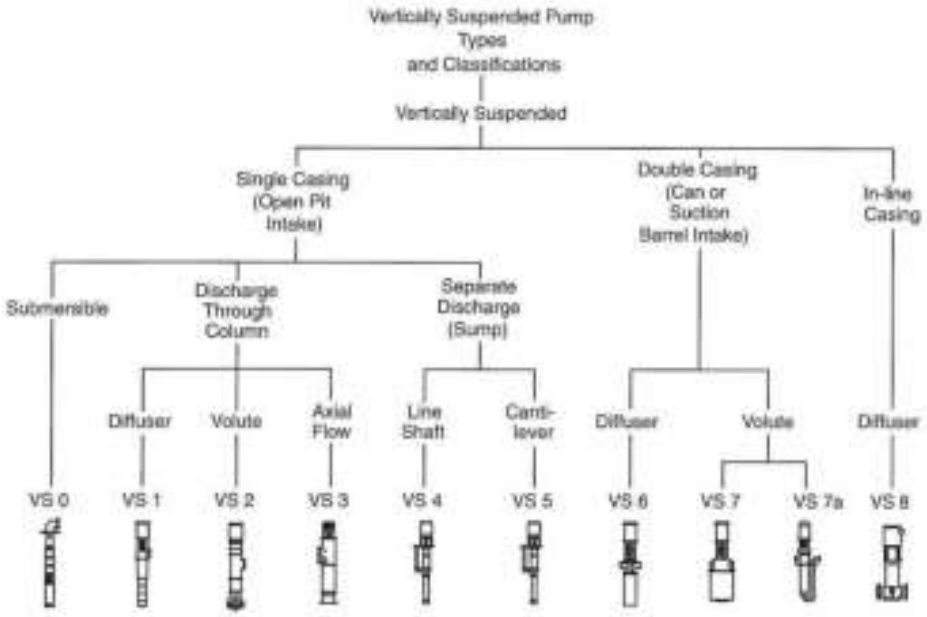

Figure 11 : Vertically suspended pumps

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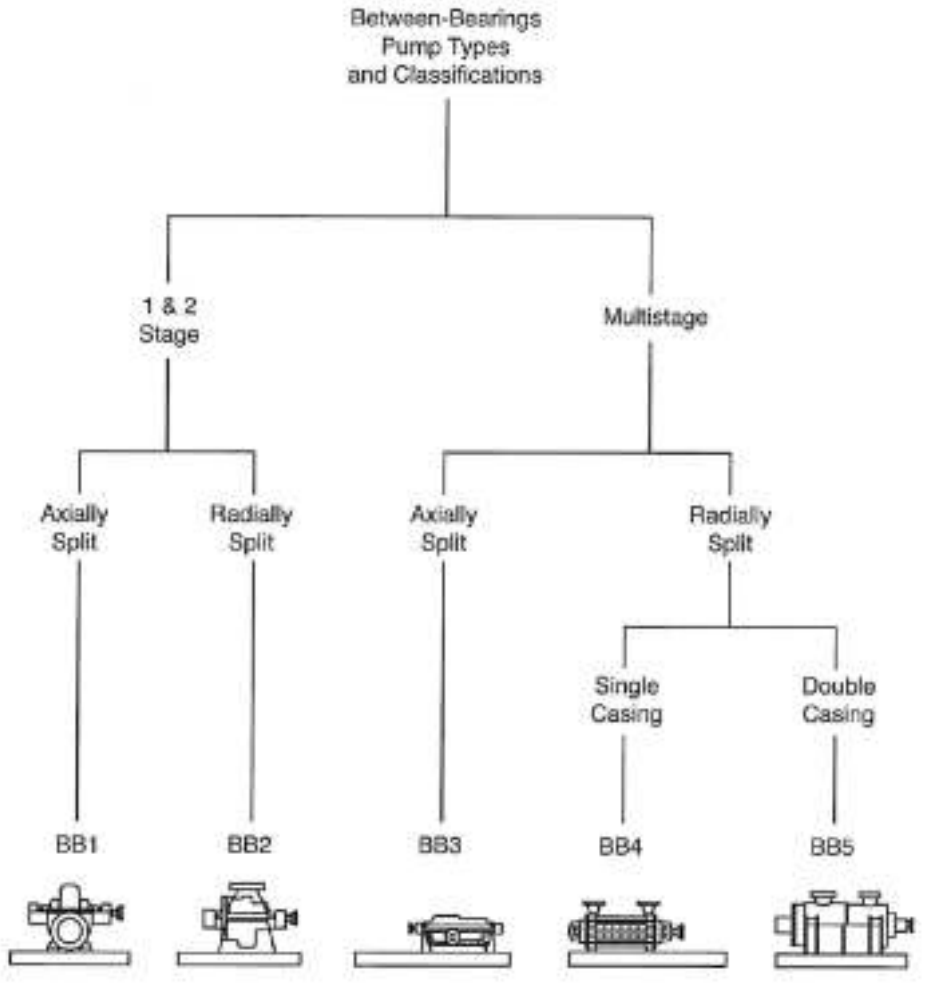

Figure 12 : Between bearings pumps

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